Predictive Analytics for Smart City Infrastructure

M.Tech Project Work Final Review

Presented by

S K Sricharan (25MML1010)

M.Tech Computer Science and Engineering

(Artificial Intelligence and Machine Learning – In collaboration with LTI Mindtree)

Under the Guidance of

Dr R. Srinivasa Perumal

Associate Professor

School of Computer Science and Engineering

Vellore Institute of Technology, Chennai.

CONTENTS

- ABSTRACT
- INTRODUCTION
- LITERATURE SURVEY
- RESEARCH GAPS
- OBJECTIVES
- RESEARCH WORK DONE
  - MODULE IDENTIFICATION
  - SYSTEM ARCHITECTURE
  - DEVELOPMENT & INTEGRATION
  - PROTOTYPE / RESEARCH FRAMEWORK
- CONCLUSIONS
- OUTCOME
- TIMELINE / GANTT CHART
- REFERENCES

1.ABSTRACT

Traffic control rooms can see congestion as it happens, but rarely receive forecasts that are explained, honest about their uncertainty and tested before an action is suggested. This project builds a decision-support prototype on public New York City data. A context-gated GATv2–Transformer forecasts link speeds 15, 30 and 60 minutes ahead on a graph of 16 Manhattan DOT speed links (January–October 2026), fusing weather, 311 and calendar context through an availability-aware gate. Split-conformal intervals quantify uncertainty, grouped SHAP explains the trained model, and a LangGraph workflow uses a local LLM (qwen2.5 7B) with FAISS retrieval to draft signal advisories that deterministic safety rules and a SUMO simulation must pass before human review. On a 45-day test period the model reaches 4.77, 5.33 and 6.15 mph MAE, 7–16 % below persistence; intervals reach 90 % coverage overall, and retrieval removed every fabricated citation. An audit of the simulation testbed showed it cannot yet support a controller comparison, which is therefore reported as not run.

2. INTRODUCTION

- Urban congestion depends on demand, neighbouring roads, weather, events and incidents, yet control rooms mostly see only current conditions.
- Graph neural networks forecast traffic on road networks [1]–[3], but give point forecasts without uncertainty or explanation.
- An LLM can phrase advice convincingly without evidence; retrieval grounding [4] and simulation testing [5] are needed before any signal action.
- Objective: join forecasts, uncertainty, explanations, cited evidence and simulation checks in one auditable, human-in-the-loop advisory pipeline on real NYC data.

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[5] P. A. Lopez et al., “Microscopic traffic simulation using SUMO,” in Proc. IEEE ITSC, 2018, pp. 2575–2582.

3.LITERATURE SURVEY

S.No	Article Title	Techniques Used	Key Findings	Limitations
1	[1] Li et al., Diffusion Convolutional Recurrent Neural Network (ICLR 2018)	Diffusion convolution on a directed road graph inside a GRU encoder–decoder (seq2seq)	Modelling traffic as diffusion on the directed graph improved multi-step speed forecasts over non-graph baselines	Fixed, predefined graph; point forecasts only; no exogenous context or uncertainty
2	[2] Yu et al., Spatio-Temporal Graph Convolutional Networks (IJCAI 2018)	Graph convolution combined with gated temporal convolution; fully convolutional, no recurrence	Accurate short-term forecasts with faster training and fewer parameters than recurrent models	Static graph; point forecasts; no weather or event context
3	[3] Wu et al., Graph WaveNet (IJCAI 2019)	Self-adaptive adjacency matrix learned from data plus dilated causal convolutions	Captures hidden spatial dependencies even when the given graph is incomplete; long receptive field	Learned graph is hard to interpret; no uncertainty estimates
4	[6] Brody et al., How Attentive are Graph Attention Networks? (ICLR 2022)	GATv2: reorders the attention computation so that attention becomes dynamic	Shows that GAT attention is static; GATv2 is strictly more expressive and performs better on benchmarks	General-purpose graph layer; no temporal modelling

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[6] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?,” in Proc. ICLR, 2022.

3.LITERATURE SURVEY (CONTD.)

S.No	Article Title	Techniques Used	Key Findings	Limitations
5	[7] Angelopoulos & Bates, A Gentle Introduction to Conformal Prediction (2021)	Split conformal prediction wrapped around any trained model	Distribution-free, finite-sample marginal coverage guarantee	Assumes exchangeable data; guarantees marginal, not conditional, coverage
6	[8] Xu & Xie, Conformal Prediction Interval for Dynamic Time-Series (ICML 2021)	EnbPI: bootstrap ensemble with conformal-style intervals for time series	Approximately valid intervals without exchangeability, under stated assumptions on the errors	Requires an ensemble of models; validity is approximate
7	[9] Lundberg & Lee, A Unified Approach to Interpreting Model Predictions (NeurIPS 2017)	SHAP: Shapley-value feature attribution (KernelSHAP and model-specific variants)	Unifies additive attribution methods with consistency guarantees	Cost grows with the number of features; attributions describe the model, not causes
8	[4] Lewis et al., Retrieval-Augmented Generation (NeurIPS 2020)	Dense passage retriever combined with a seq2seq generator	More specific and factual generation on knowledge-intensive tasks than a parametric-only model	Bounded by corpus and retriever quality; can still produce unsupported text

[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[7] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
[8] C. Xu and Y. Xie, “Conformal prediction interval for dynamic time-series,” in Proc. ICML, PMLR vol. 139, 2021, pp. 11559–11569.
[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Proc. NeurIPS, 2017.

4. RESEARCH GAPS

G1 Forecast stops at the number

Graph forecasting studies [1]–[3] report accuracy but do not connect the forecast to a checked, testable action.

G2 Uncertainty does not decide

Point forecasts dominate; interval width and data quality rarely decide whether an advisory is issued.

G3 Surrogate explanations

Explanations are often computed on simpler surrogate models instead of the deployed forecaster.

G4 Ungrounded LLM advice

LLM recommendations are rarely tied to cited, allow-listed evidence or bounded by deterministic safety rules.

G5 Missing context is hidden

Dead sensors and unavailable feeds are usually imputed silently instead of being modelled with availability masks.

G6 Untested testbeds

Simulation studies seldom audit route choice, demand and signal timing before comparing controllers.

5. OBJECTIVES

O1 Forecast link speeds 15, 30 and 60 min ahead with a context-gated GATv2–Transformer and compare it with six baselines on identical data

Addresses: G1, G5

O2 Quantify uncertainty with per-horizon split-conformal intervals and defer advisories when reliability is low

Addresses: G2

O3 Explain the trained model itself with grouped SHAP and sensitivity tests

Addresses: G3

O4 Draft advisories with a local LLM that must cite retrieved evidence and pass deterministic safety checks

Addresses: G4

O5 Test admissible advisories in a SUMO simulation, after verifying the testbed itself

Addresses: G6

O6 Make every result reproducible and traceable: tests, MLflow, Docker, evidence registry and a paper generated from artifacts

Addresses: all

6. RESEARCH WORK DONE

- The research work is organised in four parts:
  - ❖ **MODULE IDENTIFICATION**
  - ❖ **SYSTEM ARCHITECTURE**
  - ❖ **DEVELOPMENT & INTEGRATION**
  - ❖ **PROTOTYPE / RESEARCH FRAMEWORK**

1,968,303
raw speed records processed

16
Manhattan links modelled

7
forecasting models compared

66
automated tests passing

MODULE IDENTIFICATION

M1 Data ingestion
NYC Open Data, Open-Meteo, OSM; resumable downloads with SHA-256 provenance

M2 Preprocessing & features
5-minute grid, causal lags, availability masks, leakage-tested splits

M3 Road graph
Directed DOT-link graph (26 nodes, 16 edges) as a PyG Temporal signal [10]

M4 Forecasting
Context-gated GATv2–Transformer plus six baselines at 15/30/60 min

M5 Uncertainty
Per-horizon split-conformal intervals and a reliability gate

M6 Explainability
Grouped KernelSHAP on the trained model; neighbour and window sensitivity

M7 Knowledge & RAG
15 approved documents, MiniLM embeddings, FAISS index

M8 LLM agent
LangGraph workflow; local qwen2.5 7B via Ollama; schema-validated JSON

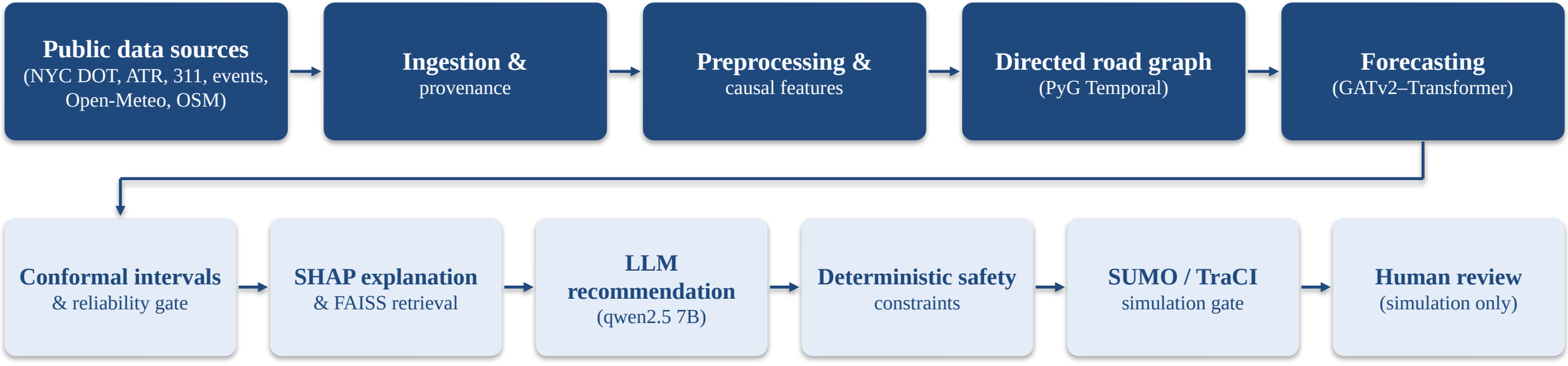
M9 Safety constraints
13 deterministic checks; protected TraCI signal controller

M10 Simulation
SUMO 1.28 West Street corridor; matched seeds; testbed audit

M11 Services
FastAPI, PostgreSQL/PostGIS job queue, worker, Streamlit dashboard

M12 Tracking & reporting
MLflow, evidence registry, claims ledger, paper tables from artifacts

SYSTEM ARCHITECTURE



Data flow

Raw API responses stored verbatim with hashes → 5-minute grid per link → windows of 24 steps × nodes × features → forecasts with 90 % intervals → advisory state in LangGraph → decision and audit trail persisted in PostgreSQL/PostGIS; every run logged in MLflow.

Technology stack

Python 3.12 · Pandas/Polars · PyTorch + PyTorch Geometric + PyG Temporal · XGBoost/LightGBM · SHAP · sentence-transformers + FAISS · Ollama (qwen2.5:7b) · LangGraph · SUMO 1.28 / TraCI · FastAPI · Streamlit · PostgreSQL/PostGIS · MLflow · Docker Compose · Pytest

DEVELOPMENT & INTEGRATION

Implementation of modules

One Python package with a staged command-line interface, YAML configuration and Pydantic contracts

Every stage writes artifacts with run IDs and SHA-256 hashes

The same code path runs the smoke tests and the full experiment

Integration strategy

Shared contracts frozen first: forecast, uncertainty, evidence, candidate action, decision

A stage runner executes the real-data pipeline in fixed order with a stage log

Only the recommendation node calls the LLM; every gate is deterministic code

66 automated tests: unit, leakage, safety, integration, end-to-end

Challenges faced → solutions

7 of 26 links report zero speed all year
→ treated as missing; 16 links kept by a train-coverage rule

DOT link geometries stop short of link ends
→ three documented graph-edge rules

GRU crash with cuDNN on Windows/CUDA
→ cuDNN disabled for GRU runs only

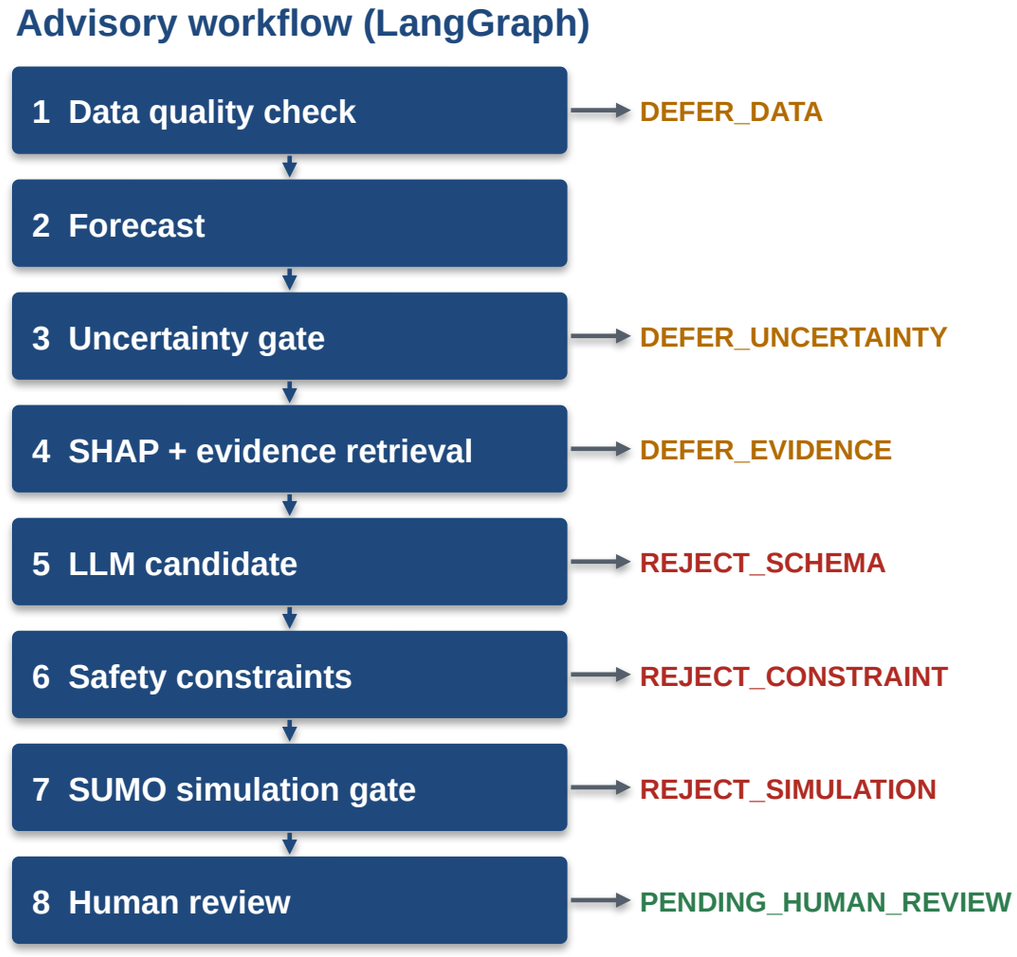
Bootstrap resampled 5-minute steps, not days
→ unit fixed and ablations re-run

A missing field silently blocked every LLM call
→ fixed and agent evaluation re-run

SUMO re-routed half the corridor traffic
→ fixed routes and a testbed audit

PROTOTYPE / RESEARCH FRAMEWORK

- **Command-line pipeline:** one command per stage, from data audit to paper tables
- **FastAPI service:** 18 endpoints for forecasts, explanations, retrieval, advisories, simulations and review; long jobs run in a PostgreSQL-backed worker
- **Streamlit dashboard:** 11 pages, from data audit and traffic map to forecasts, explanations, agent traces, SUMO results and the model card
- **Every view labels its mode:** LIVE, REPLAY or SIMULATION
- **Docker Compose:** PostGIS, API, worker and dashboard images, built and smoke-tested
- **Human review** approves or rejects for simulation only; nothing is sent to real signals



7. CONCLUSIONS

Performance metrics (test period 6 Sep – 20 Oct 2026)

Metric	15 min	30 min	60 min
Proposed model MAE (mph)	**4.77**	**5.33**	**6.15**
Proposed, mean of 3 seeds (mph)	4.69	5.34	6.15
Persistence MAE (mph)	5.14	6.06	7.29
90 % interval coverage	0.909	0.907	0.903
Coverage under congestion	0.798	0.794	0.757
MAE kept after gate (≈26 % deferred)	3.72	4.38	5.25

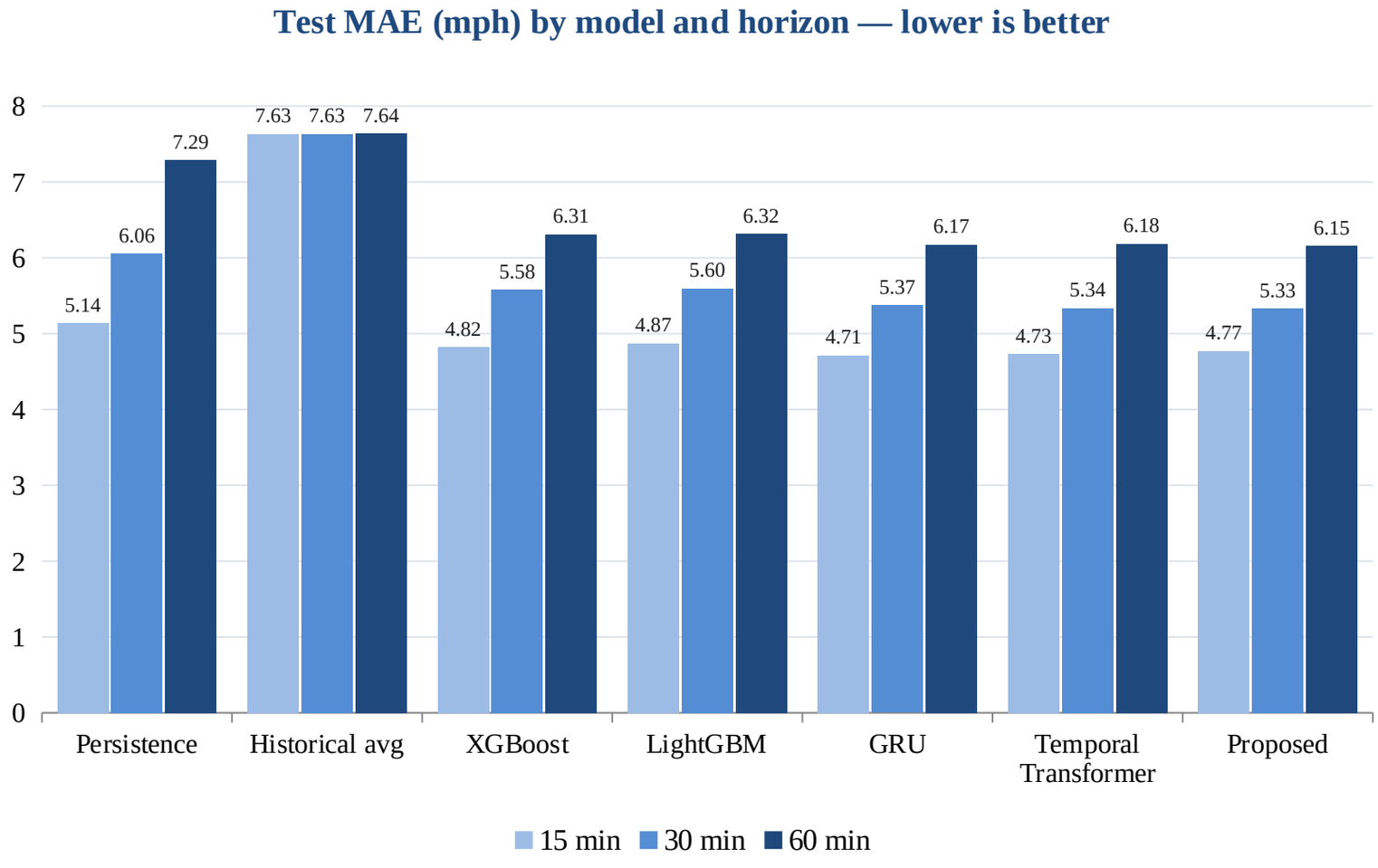
Validation of approach

Chronological splits with purge gaps and leakage tests; identical targets and masks for every model; day-block bootstrap for ablations; a preregistered simulation baseline test; failed runs kept and reported.

Observations

- The proposed model beats persistence by 7–16 % and both tree models. Over three seeds it is 0.03–0.08 mph below the temporal Transformer at every horizon and the GRU at 30/60 min; the GRU is better at 15 min. Small, not decisive.
- The availability-aware gate helps (MAE +0.067 mph when removed); operational weather features hurt (−0.046 mph), a negative result.
- Intervals meet 90 % overall but under-cover congestion, the regime that matters for advisories.
- Retrieval removed fabricated citations (3 of 3 without it, 0 of 9 with it); the constraint layer, not the LLM, decides admissibility.
- The simulation testbed fails its own baseline test, so the controller comparison is not run.

8. OUTCOME



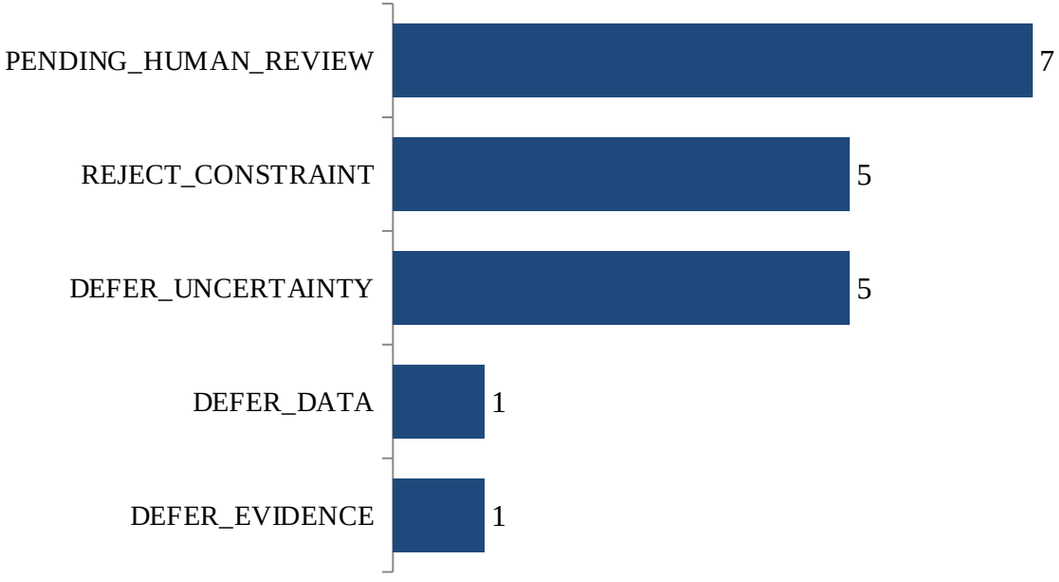
Key results

- Every learned model beats persistence and the historical average.
- Proposed model: 4.77 / 5.33 / 6.15 mph; lowest at 30 and 60 min, GRU lowest at 15 min.
- R^2 rises from 0.55 / 0.46 / 0.30 (persistence) to 0.69 / 0.63 / 0.56.
- 144,205–144,217 test targets per horizon, identical for every model.

Data: NYC DOT speeds [11], 1 Jan – 20 Oct 2026.

8. OUTCOME (CONTD.)

Advisory decisions in 19 workflow runs (REPLAY cases)



12 of 12 LLM responses were schema-valid. Without retrieval the model cited non-existent evidence in 3 of 3 cases (all rejected); with retrieval in 0 of 9.

SUMO testbed outcome (simulation only)

Protocol	Unfinished*	Result
First comparison, 20 Oct (45 runs)	—	**Withdrawn**
2014 count, generated signal timing	17.7 %	**Fails test**
2014 count, Webster re-timed signals	4.3 %	**Fails test**
2025 counts, Webster, no turn-lane limits	10.1 %	**Fails test**

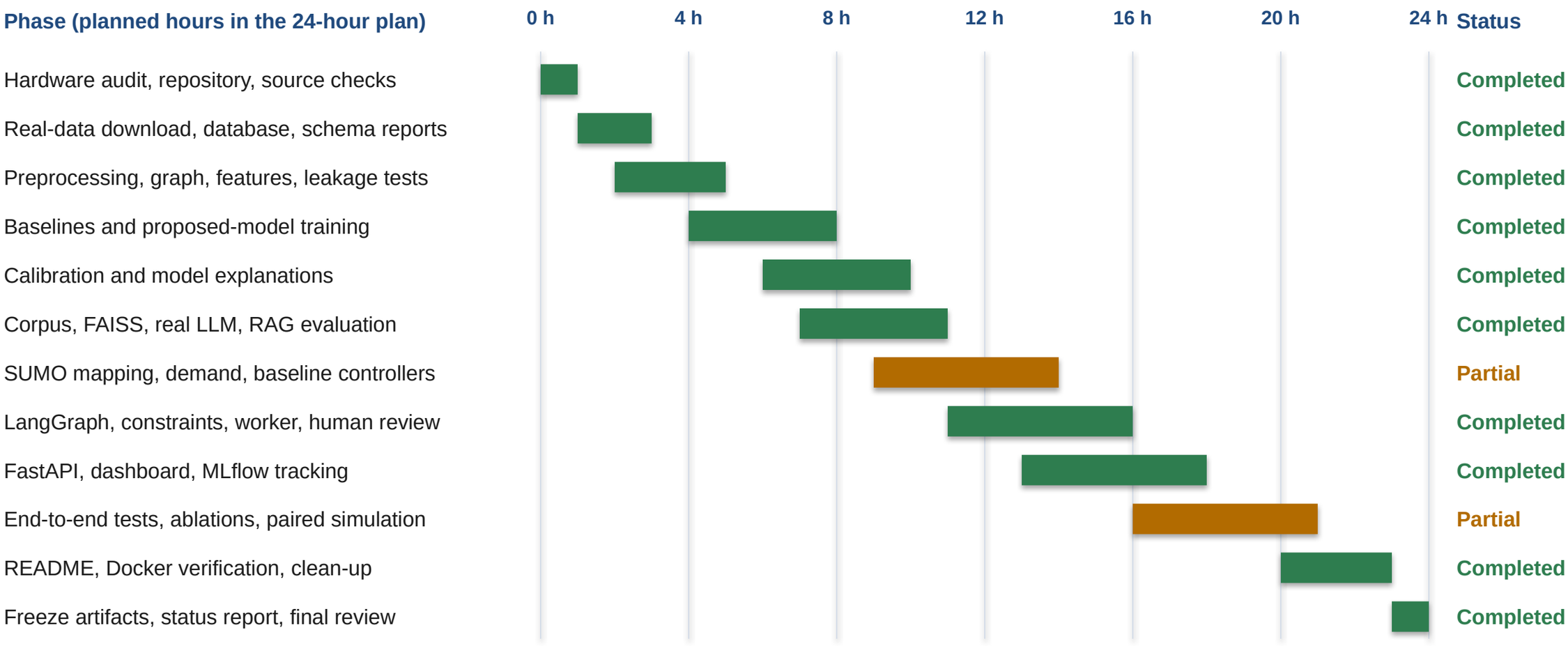
* Vehicles not finishing with no side-street traffic; the preregistered test allows at most 1 % and no stuck vehicles.

The first comparison let SUMO re-route 1,120 of 2,189 main-road vehicles off West Street, so it did not test the corridor.

With real 2025 counts (2,510 NB / 2,444 SB veh/h) the modelled link runs at 5.8 mph; the real street runs at 10.95 mph in the PM peak.

The test was not relaxed: the comparison is reported as not run, and 10.95 mph is now the calibration target.

9. TIMELINE / GANTT CHART



Bars: planned window (project specification). Colour: actual status, 19–21 Oct 2026 (repository history). Next milestones: calibrate West St signal timing to the observed 10.95 mph (or obtain DOT timing plans) → controller comparison; more seeds without the compute budget; paper submission.

REFERENCES

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.

[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.

[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.

[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.

[5] P. A. Lopez et al., “Microscopic traffic simulation using SUMO,” in Proc. IEEE ITSC, 2018, pp. 2575–2582.

[6] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?,” in Proc. ICLR, 2022.

[7] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.

[8] C. Xu and Y. Xie, “Conformal prediction interval for dynamic time-series,” in Proc. ICML, PMLR vol. 139, 2021, pp. 11559–11569.

[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Proc. NeurIPS, 2017.

[10] B. Rozemberczki et al., “PyTorch Geometric Temporal: Spatiotemporal signal processing with neural machine learning models,” in Proc. ACM CIKM, 2021, pp. 4564–4573.

[11] NYC Department of Transportation, “DOT Traffic Speeds NBE,” NYC Open Data, dataset i4gi-tjb9, accessed Oct. 2026.

[12] NYC Department of Transportation, “Automated Traffic Volume Counts,” NYC Open Data, dataset 7ym2-wayt, accessed Oct. 2026.

Thank You